

**K S Tharunpranav 251801359-2025-AIDS** ▾**K2****Started on** Thursday, 24 September 2026, 9:16 AM**State** Finished**Completed on** Thursday, 1 October 2026, 8:31 AM**Time taken** 6 days 23 hours**Marks** 1.00/1.00**Grade** **10.00** out of 10.00 (**100%**)

Question 1 | Correct Mark 1.00 out of 1.00

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer: (penalty regime: 0 %)

```

1 #include <stdio.h>
2 #include <stdlib.h>
3
4 void findIntersection() {
5     int n1, n2;
6     if (scanf("%d", &n1) != 1) return;
7     int *arr1 = (int *)malloc(n1 * sizeof(int));
8     for (int i = 0; i < n1; i++) {
9         scanf("%d", &arr1[i]);
10    }
11    if (scanf("%d", &n2) != 1) {
12        free(arr1);
13        return;
14    }
15    int *arr2 = (int *)malloc(n2 * sizeof(int));
16    for (int i = 0; i < n2; i++) {
17        scanf("%d", &arr2[i]);
18    }
19
20    int i = 0, j = 0;
21    int first = 1;
22    while (i < n1 && j < n2) {

```

```
22 // ...
23     if (arr1[i] < arr2[j]) {
24         i++;
25     } else if (arr1[i] > arr2[j]) {
26         j++;
27     } else {
28         if (!first) {
29             printf(" ");
30         }
31         printf("%d", arr1[i]);
32         first = 0;
33         i++;
34         j++;
35     }
36 }
37 printf("\n");
38 free(arr1);
39 free(arr2);
40 }
41
42 int main() {
43     int t;
44     if (scanf("%d", &t) == 1) {
45         while (t--) {
46             findIntersection();
47         }
48     }
49     return 0;
50 }
51
```

	Input	Expected	Got	
✓	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✓
✓	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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